

Mohid Tahir

Mechanical Engineer

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Education

University of Toronto

Sep. 2021 – Apr. 2026

Bachelor of Applied Science and Engineering

Toronto, ON

- Program: Mechanical Engineering + PEY Co-op, Minor: Engineering Business, Certificate: EV Design
- Related Coursework: Mechanical Engineering Design, Kinematics and Dynamics of Machines, Electric Vehicle Design, Analog and Digital Electronics, Design for the Environment

Meadowvale Secondary School

Sep. 2017 – Jun. 2021

Ontario Secondary School Diploma

Mississauga, ON

Technical Skills and Qualifications

Software & Tools: SolidWorks, AutoCAD, Onshape, Minitab, Microsoft Office Suite (Excel, Word, PowerPoint), Calibre

Programming & Languages: Python, MATLAB, CSS, HTML, Git, PLC, G-code

Simulation & Analysis: Life Cycle Assessment (LCA), Net Present Value (NPV) Modelling, DMAIC, Weighted Decision Matrices, Circuit Simulation (SuperSpice)

Fabrication & CAD Practices: Manual Machining (lathe, mill, drill press), PCB Design, 3D Modelling, Reverse Engineering, Tolerance Analysis

Soft Skills: Time Management, Collaboration, Critical Analysis, Verbal & Written Communication, Technical Documentation, Client Communication, Project Management

Professional Experience

Mechanical Engineering Intern (Special Projects)

May 2024 – Aug. 2025

AGS Automotive Systems

Oshawa, ON

- Designed 20+ part and process models in SolidWorks, improving manufacturing efficiency and drawing accuracy
- Updated schematic drawings in AutoCAD to ensure consistency across production projects
- Analyzed time data using the MOST Technique, achieving a 15% reduction in process times for key workflows
- Conducted quality audits on over 50 steel bundles per week, ensuring compliance with company standards
- Performed process improvement analysis in Excel, supporting workflow enhancements and data-driven decision making

Design Engineering Intern

May 2023 – Aug. 2023

Parts Apart

Toronto, ON

- Conducted the end-to-end development of a product, overseeing 4 stages of the product development lifecycle
- Performed extensive market research to identify and define a target niche, one that impacts 40% of the population
- Employed computer-aided design (CAD) and manufacturing processes to develop and enhance prototypes
- Collaborated closely with the company founder to transform the concept into a tangible product, costing under \$50

Communications Manager & Project Team Member

Jan. 2022 – Apr. 2022

Engineering Strategies and Practice II (APS112)

Toronto, ON

- Collaborated with a university professor to design a two-variable teaching device that allowed beginner engineering students to run physical DMAIC experiments
- Utilized functional decomposition, morphological charts, and multivoting to develop an adjustable ramp and continuous ball drop mechanism constrained by classroom safety limits
- Validated the system by running 25 tests across 5 variable combinations, modelling the transfer function in Minitab to prove experimental reliability

Shift Leader

Jun. 2020 – Jan. 2024

Dairy Queen

Mississauga, ON

- Led a 6 to 8 person team through weekday rushes, owning cash reconciliation, order accuracy, and escalated customer issues across 3.5 years of part-time work alongside university
- Trained new hires on POS, food safety, and service standards, shortening ramp-up time for each new cohort
- Managed end-of-shift store operations, including register balancing and daily inventory management, ensuring smooth handoffs between shifts
- Actively contributed to new product ideation and promotional rollouts while building team morale in a fast-paced environment

Projects

Low-Speed PMSG for Vertical Axis Wind Turbine

Sep. 2025 – Apr. 2026

Capstone Design (MIE490/491), Client: UTWind

University of Toronto

- Designed a custom direct-drive outer-rotor radial-flux permanent magnet synchronous generator for UTWind's 2026 vertical axis wind turbine, targeting 600 to 1200 W output at 100 to 200 RPM with under 0.1 Nm starting torque
- Screened four generator architectures using a weighted decision matrix, eliminating candidates with poor low-RPM efficiency and high cogging torque
- Collaborated with teammates to optimize rotor and stator geometry based on electromagnetic simulation data, successfully hitting 842 W output and 0.09 Nm peak starting torque
- Delivered a fabrication-ready SolidWorks assembly with procurement specs for NdFe35 magnets and ASTM A677 (M19) lamination steel, setting up the 2026 turbine build

The WindRunner

Jan. 2026 – Apr. 2026

Product Design (MIE540)

University of Toronto

- Designed a propeller-driven sprint car targeting a \$15 retail price and a strict \$6 BOM, balancing speed, durability, and a child-safe mesh guard constraint
- Conducted full factorial DOE and factor analysis, identifying an 85 mm wheel diameter as the dominant variable that improved track times by nearly 10 seconds across carpet and tile surfaces
- Performed worst-case and RSS tolerance analysis to finalize component geometry, validating the optimized physical prototype against Voice of Customer requirements

All-Wheel-Drive Demonstration Platform

Sep. 2025 – Dec. 2025

Electric Vehicle Design (APS380)

University of Toronto

- Designed and built a scaled dual-motor split-axle AWD platform in SolidWorks, integrating front and rear differentials to demonstrate EV drivetrain architecture and torque distribution
- Instrumented the platform with an ESP32 microcontroller and Hall-effect sensors to capture wheel-speed data and observe traction slip behavior during live test runs
- Iterated on belt and pulley geometry across multiple CAD revisions, resolving critical belt-skip issues and adjusting to 3D printing tolerances after pivoting from an initial planetary gear layout

Printed Circuit Board

Jan. 2024 – Apr. 2024

Analog & Digital Electronics (MIE346)

University of Toronto

- Designed a custom printed circuit board to scale a low-voltage analog signal from 0 to 0.35 V to a 0.25 to 3.3 V microcontroller range, incorporating a non-inverting op-amp configuration
- Developed a two-stage Butterworth low-pass filter with a 1 kHz cutoff using the Analog Devices Filter Tool, simulating circuit sweeps in SuperSpice
- Executed schematic capture and PCB layout using Multisim, Ultiboard, and Eagle, successfully fabricating the board and validating less than 4% error against theoretical predictions

Google Home Housing

Jan. 2024 – Apr. 2024

Design for the Environment (MIE315)

University of Toronto

- Ran a materials trade study comparing plastic and FSC-certified wood for smart speaker housings, utilizing a weighted decision matrix and Life Cycle Assessment (LCA)
- Modelled a 5-year Net Present Value (NPV) at a 5% rate, demonstrating wood achieved a more favourable NPV (-\$82.10 vs -\$178.34) alongside lower cumulative emissions
- Delivered a comprehensive consulting recommendation that owned the engineering trade-offs, factoring in machining energy, varnish ecotoxicity, acoustics, and public perception

Robotic Arm for Rover

Sep. 2023 – Dec. 2023

Kinematics and Dynamics of Machines (MIE301)

University of Toronto

- Redesigned a sample-collection robotic arm for Io's terrain, adapting the architecture from a rover base to a compact drone-mounted configuration
- Optimized link lengths for a folding four-bar linkage using MATLAB, ensuring the entire assembly packed strictly within a 58.5 cm by 48.9 cm drone envelope
- Modelled the joint stack and frame in SolidWorks, selecting Aluminum Alloy 1060 to maximize the strength-to-weight ratio for aerial deployment

Pick and Place CNC Machine

Jan. 2023 – Apr. 2023

Mechanical Engineering Design (MIE243)

University of Toronto

- Conceptualized a desktop CNC pick-and-place machine for educational PCB assembly, focusing on modularity, affordability, and a ± 0.01 inch placement accuracy
- Designed a 3-axis motion system in SolidWorks, integrating a lead screw z-axis with sleeve bearings, interchangeable rotational vacuum heads, and a custom ratchet-indexed reel feed
- Sourced all mechanical components through McMaster-Carr to generate a realistic BOM, and specified a camera-based alignment subsystem to ensure reliable part orientation

Mechanical Dissections

Jan. 2023 – Apr. 2023

Mechanical Engineering Design (MIE243)

University of Toronto

- Reverse-engineered physical devices including a jigsaw, can opener, and screwdriver, utilizing calipers to measure parts and rebuild them digitally in SolidWorks
- modelled a complete 12:1 parallel gearbox from scratch, correctly dimensioning and mating gears, shafts, and housings to simulate functional rotation
- Produced hand-drafted engineering drawings and detailed parametric models, bridging the gap between physical measurement and CAD assembly

Miniature Pneumatic Engine

Nov. 2022

George Brown College Machining

Toronto, ON

- Machined a functional pneumatic engine using manual mills, lathes, and drill presses, applying foundational fabrication skills
- Optimized feed rates and cutting speeds to ensure tight dimensional tolerances, safe operation, and high-quality surface finishes

Short Story – “Snow Falling Soundlessly at Night”

Feb. 2020

Meadowvale Secondary School

Mississauga, ON

- Authored and formatted an original short story as an EPUB publication for a scholarship submission
- Explored narrative themes of memory and resilience to showcase strong written communication and structured project execution

Handmade Coffee Table & Jewellery Box

Dec. 2018

Meadowvale Secondary School

Mississauga, ON

- Designed and assembled functional wooden furniture pieces, utilizing band saws and belt sanders to achieve precise symmetry
- Applied professional finishing techniques and dimensional planning to translate raw materials into project-ready pieces

CO₂ Gas Car & Birdhouse Construction

Oct. 2017 – Dec. 2017

Meadowvale Secondary School

Mississauga, ON

- Constructed a functional birdhouse and a CO₂-powered wooden car, applying early measurement and cutting skills
- Utilized foundational engineering principles including aerodynamics, weight distribution, and structural layout to optimize design performance

Volunteering & Extra-Curriculars

Frosh Leader – Skule Frosh Team

Sep. 2022

University of Toronto

Toronto, ON

- Led orientation activities for new engineering students, promoting inclusion and engagement across diverse cohorts

Photographer

Nov. 2019 – Mar. 2020

Meadowvale Secondary School

Mississauga, ON

- Captured key school and community events for newspaper publication, using photography to document student life

Summer Camp Counsellor

Aug. 2019

City of Mississauga

Mississauga, ON

- Led orientation activities for youth, promoting inclusion and engagement across diverse cohorts

World of Welcome Volunteer

Aug. 2019

Peel District School Board

Mississauga, ON

- Helped immigrant students acclimate to school life in Canada by leading orientation and peer support activities

Event Assistant

Jul. 2019

ICNA Relief Canada

Mississauga, ON

- Supported event logistics and merchandise sales for charity events, ensuring smooth setup and breakdown

Awards & Certifications

Dean's Honours List, Certified SolidWorks Associate (CSWA), Basic Machining (George Brown), Secondary School Valedictorian, Ontario Scholar, Subject Award – International Business (99%), Subject Award – Canadian Law (99%)